Semana 6: Raízes n-ésimas dos Complexos

$z = a + bi$

$$w = \sqrt[n]{z} \Leftrightarrow w^n = z$$

$z = \rho(\cos\theta + i\sin\theta)$

$w = r(\cos\varphi + i\sin\varphi) \Longrightarrow w^n = r^n(\cos(n\varphi) + i\sin(n\varphi))$

$$w^n = z \Leftrightarrow r^n(\cos(n\varphi) + i\sin(n\varphi)) = \rho(\cos\theta + i\sin\theta)$$

$r = \sqrt[n]{\rho}$

$n\varphi = \theta + 2k\pi \Longrightarrow \varphi = \dfrac{\theta + 2k\pi}{n}$

$$w = \sqrt[n]{\rho}\left(\cos\left(\frac{\theta + 2k\pi}{n}\right) + i\sin\left(\frac{\theta + 2k\pi}{n}\right)\right)$$

$$k = 0 \Longrightarrow \varphi = \frac{\theta}{n}$$

$$\vdots$$

$$\varphi = \frac{\theta + 2k\pi}{n} = \frac{\theta}{n} + \frac{2k\pi}{n}$$

$$k = n - 1 \Longrightarrow \varphi = \frac{\theta}{n} + \frac{2(n-1)\pi}{n}$$

$$k = n \Longrightarrow \varphi = \frac{\theta}{n} + \frac{2n\pi}{n} = \frac{\theta}{n} + 2\pi$$

$$\sqrt[n]{\boldsymbol{z}} = \sqrt[n]{\boldsymbol{\rho}}\left(\boldsymbol{\cos}\left(\frac{\boldsymbol{\theta + 2k\pi}}{\boldsymbol{n}}\right) + \boldsymbol{i}\,\boldsymbol{\sin}\left(\frac{\boldsymbol{\theta + 2k\pi}}{\boldsymbol{n}}\right)\right) \qquad \boldsymbol{k = 0, 1, 2, \ldots, n - 1}$$

$$\sqrt[4]{-8+8\sqrt{3}i} \qquad z=-8+8\sqrt{3}i$$

$$\rho=|z|=\sqrt{(-8)^2+\left(8\sqrt{3}\right)^2}=\sqrt{256}=16$$

$$\begin{cases}\cos\theta=\frac{Re(z)}{|z|}=\frac{-8}{16}=-\frac{1}{2}\\ \sin\theta=\frac{Im(z)}{|z|}=\frac{8\sqrt{3}}{16}=\frac{\sqrt{3}}{2}\end{cases}\Longrightarrow\theta=\frac{2\pi}{3}$$

$$z=16\left(cos\left(\frac{2\pi}{3}\right)+i\sin\left(\frac{2\pi}{3}\right)\right)$$

$$\sqrt[4]{z} = \sqrt[4]{16}\left(cos\left(\frac{\frac{2\pi}{3}+2k\pi}{4}\right)+i\sin\left(\frac{\frac{2\pi}{3}+2k\pi}{4}\right)\right) \qquad k=0,1,2,3$$

$$\sqrt[4]{z} = 2\left(cos\left(\frac{\pi}{6}+\frac{k\pi}{2}\right)+i\sin\left(\frac{\pi}{6}+\frac{k\pi}{2}\right)\right) \qquad k=0,1,2,3$$

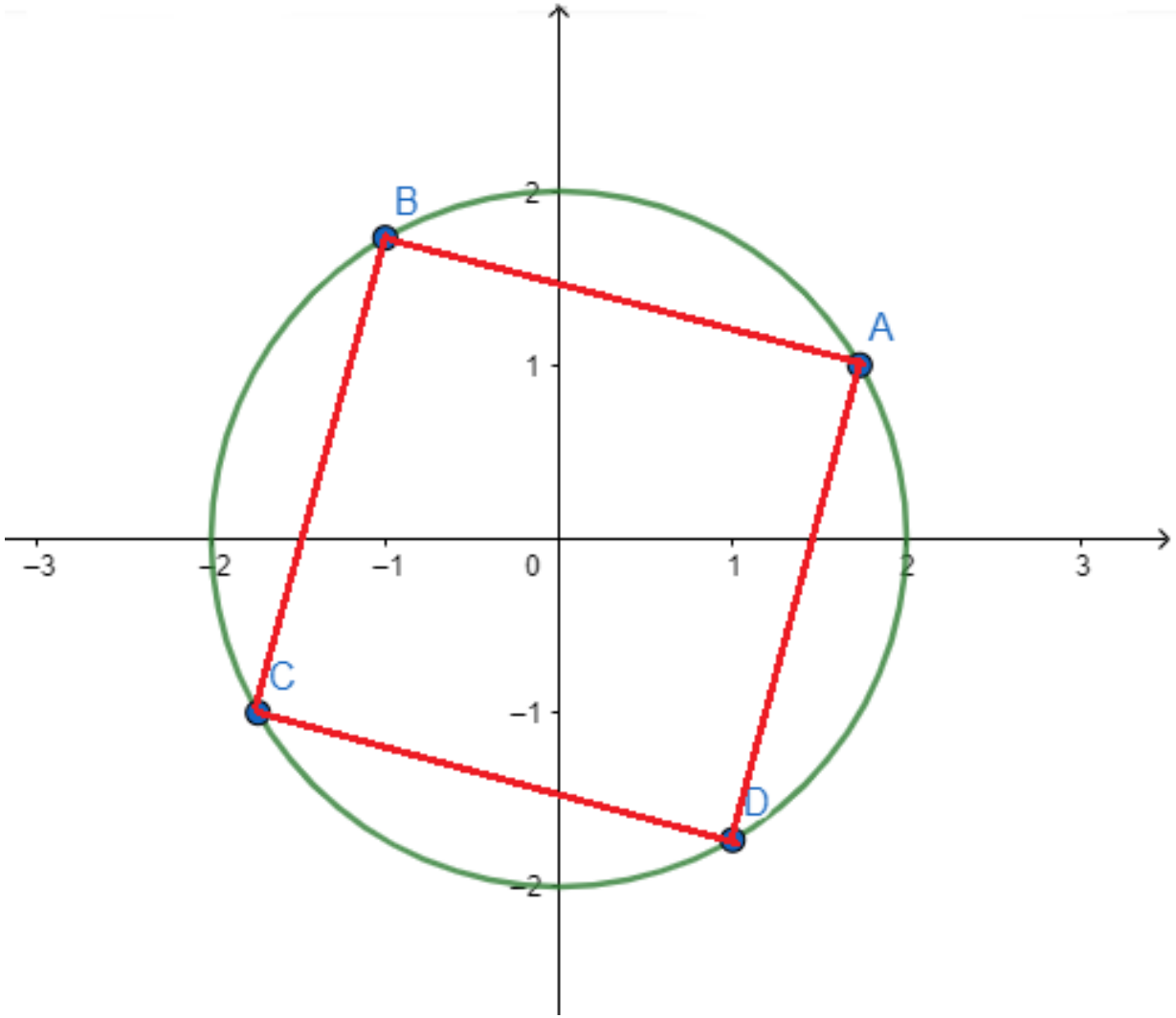
$$k=0 \Longrightarrow 2\left(cos\left(\frac{\pi}{6}\right)+i\sin\left(\frac{\pi}{6}\right)\right)=2\left(\frac{\sqrt{3}}{2}+\frac{1}{2}i\right)=\sqrt{3}+i$$

$$k=1 \Longrightarrow 2\left(cos\left(\frac{\pi}{6}+\frac{\pi}{2}\right)+i\sin\left(\frac{\pi}{6}+\frac{\pi}{2}\right)\right)=2\left(cos\left(\frac{2\pi}{3}\right)+i\sin\left(\frac{2\pi}{3}\right)\right)=2\left(-\frac{1}{2}+\frac{\sqrt{3}}{2}i\right)=-1+\sqrt{3}i$$

$$k=2 \Longrightarrow 2\left(cos\left(\frac{\pi}{6}+\frac{2\pi}{2}\right)+i\sin\left(\frac{\pi}{6}+\frac{2\pi}{2}\right)\right)=2\left(cos\left(\frac{7\pi}{6}\right)+i\sin\left(\frac{7\pi}{6}\right)\right)=2\left(-\frac{\sqrt{3}}{2}-\frac{1}{2}i\right)=-\sqrt{3}-i$$

$$k=3 \Longrightarrow 2\left(cos\left(\frac{\pi}{6}+\frac{3\pi}{2}\right)+i\sin\left(\frac{\pi}{6}+\frac{3\pi}{2}\right)\right)=2\left(cos\left(\frac{5\pi}{3}\right)+i\sin\left(\frac{5\pi}{3}\right)\right)=2\left(\frac{1}{2}-\frac{\sqrt{3}}{2}i\right)=1-\sqrt{3}i$$

$$\sqrt[4]{z} = 2\left(cos\left(\frac{\pi}{6}+\frac{k\pi}{2}\right)+i\sin\left(\frac{\pi}{6}+\frac{k\pi}{2}\right)\right) \qquad k = 0,1,2,3$$



Observação: Método Direto

$$\sqrt{-7-24i}$$

$$\sqrt{-7-24i} = x + yi \Leftrightarrow -7-24i = x^2 - y^2 + 2xyi \Leftrightarrow \begin{cases} x^2 - y^2 = -7 \\ 2xy = -24 \end{cases}$$

Soluções do Sistema: $x = 3, y = -4$ e $x = -3, y = 4$

$$\sqrt{-7-24i} = \pm(3-4i)$$

Raízes n-ésimas da Unidade: $\sqrt[n]{1}$

$$1 = (\cos 0 + i \sin 0) \Longrightarrow \sqrt[n]{1} = \cos\left(\frac{2k\pi}{n}\right) + i \sin\left(\frac{2k\pi}{n}\right), k = 0,1,2, \dots, n-1$$

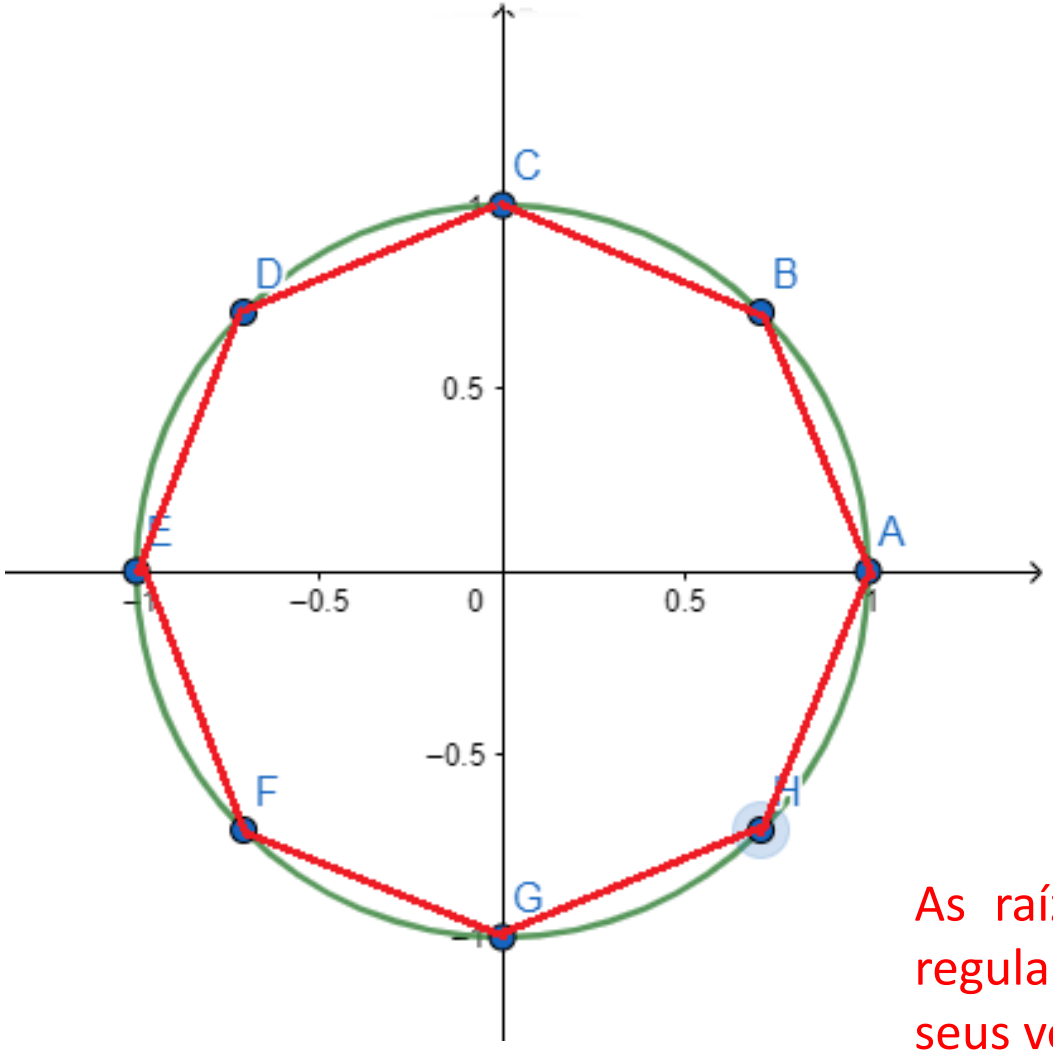
$$\omega = \cos\left(\frac{2\pi}{n}\right) + i \sin\left(\frac{2\pi}{n}\right)$$

Raízes n-ésimas da unidade: $1, \omega, \omega^2, \omega^3, \dots \omega^{n-1}$

$$\sqrt[n]{z} = \sqrt[n]{\rho}\left(cos\left(\frac{\theta + 2k\pi}{n}\right) + i \sin\left(\frac{\theta + 2k\pi}{n}\right)\right) = \sqrt[n]{\rho}\left(cos\left(\frac{\theta}{n}\right) + i \sin\left(\frac{\theta}{n}\right)\right)\left(cos\left(\frac{2k\pi}{n}\right) + i \sin\left(\frac{2k\pi}{n}\right)\right)$$

As raízes n-ésimas de um número complexo são obtidas pelo produto de uma raiz n-ésima particular com as raízes n-ésimas da unidade ω^k

$$\sqrt[8]{1} = cos\left(\frac{2k\pi}{8}\right) + i\,\sin\left(\frac{2k\pi}{8}\right) = cos\left(\frac{k\pi}{4}\right) + i\,\sin\left(\frac{k\pi}{4}\right) \qquad \mathrm{k} = 0,1,2,\ldots,7$$



Raízes oitavas da unidade:

1	$\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i$	i	$-\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i$
-1	$-\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i$	$-i$	$\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i$

As raízes n-ésimas da unidade são os vértices de um polígono regular de n lados, inscrito no círculo unitário, tendo como um de seus vértices a própria unidade $(1,0)$.

$$\sqrt[3]{\mathbf{8}} = ?$$

Raiz cúbica particular: 2

Raízes cúbicas da unidade: vértices de um triângulo equilátero inscrito no círculo unitário

$\theta = 0 \Longrightarrow 1$

$\theta = \frac{2\pi}{3} \Longrightarrow -\frac{1}{2} + \frac{\sqrt{3}}{2}i$

$\theta = \frac{4\pi}{3} \Longrightarrow -\frac{1}{2} - \frac{\sqrt{3}}{2}i$

$$\sqrt[3]{8} = \quad 2, \quad -1 + \sqrt{3}i, \quad -1 - \sqrt{3}i$$

Decomponha o polinômio $x^4 + 1$ como um produto de 2 polinômios de grau 2 e coeficientes reais

$$x^4 + 1 = 0 \Leftrightarrow x = \sqrt[4]{-1}$$

$$-1 = \cos\pi + i\sin\pi \Longrightarrow \sqrt[4]{-1} = \cos\left(\frac{\pi}{4} + \frac{k\pi}{2}\right) + i\sin\left(\frac{\pi}{4} + \frac{k\pi}{2}\right)$$

$$k = 0 \Longrightarrow \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i \qquad k = 1 \Longrightarrow -\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i \qquad k = 2 \Longrightarrow -\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i \qquad k = 3 \Longrightarrow \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i$$

$$w = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i$$

$$x^4 + 1 = (x - w)(x - \overline{w})(x + w)(x + \overline{w})$$

$$= (x^2 - x(w + \overline{w}) + |w|^2)(x^2 + x(w + \overline{w}) + |w|^2)$$

$$= \left(x^2 - \sqrt{2}x + 1\right)\left(x^2 + \sqrt{2}x + 1\right)$$